PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT EXAMINATIONS SYNDICATE

FORM TWO MOCK EXAMINATIONS AUGUST 2018

041 BASIC MATHEMATICS

Time 2:30HRS

INSTRUCTIONS

1.This paper consists of ten (10) compulsory questions
2.Show clearly all the working and answers in your answer sheets
3.All writings must be in blue or black ink except for drawings which must be in pencil
4.Four figure mathematical tables, geometric instructions and graph papers may be used where necessary
5.All communication devises and calculators are not allowed in the examinations room
6.Write your examination number on every page of your answer sheet
7.Cheating is not allowed
8.Answer all questions

1. (a) Find the LCM and GCF of $12,24 \text{ and } 36$

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(b) Write the following number 247.1387

(i) Correct to 4 significant figures

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(ii) In standard form

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(iii) To nearest hundredth

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2. (a) Convert $\frac{1}{4}$ into decimal and percentage

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(b) By using the number line perform $2-6-(-8)$

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(c) Convert $2.1\dot{3}$ into a fraction

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3. (a) Change $2\frac{1}{2}$ *metres* into centimeters

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(b) Find the time in which sh.$350{,}000$ Will earn sh. $38{,}000$ at the rate of 4% interest per annum

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(c) Estimate the value of 489×43117

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4. (a) A chair is bought for $2{,}000/=$ and then sold for $2{,}400/=$ Find

(i) The profit

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(ii) Percentage profit

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(b) Find the sum of interior angles of 14 *sided* polygon

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5. (a) Solve by using elimination method

$$\begin{cases} 2x - 3y = 5 \ldots\ldots\ldots (i) \\ 5x - 2y = 18 \ldots\ldots\ldots (ii) \end{cases}$$

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(b) Find the equation of a line joining the points $A(-2,3)$ *and* $B(4,4)$

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6. (a) Find the values of the following

(i) $(5)^3$

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(ii) $(27)^{\frac{1}{3}}$

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(iii) $(3)^{-4} \times (3)^{-8}$

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(b) Rationalize the denominator $\frac{1}{\sqrt{2}+\sqrt{3}}$

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7. (a)(i) Find $\log_{10} 1000$

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(ii)Solve the equation $\log_3(2x+1) - \log_3(x-3) = 2$

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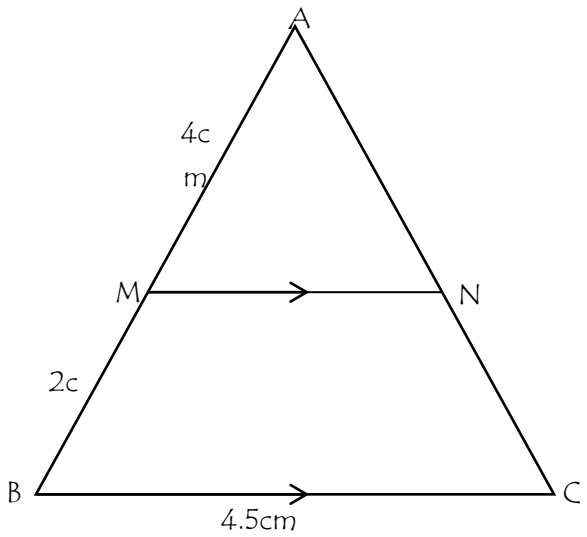
(b) By using mathematical table, Evaluate $\frac{1.663 \times 4.550}{\sqrt{201.23}}$

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8. (a) A class of students took a mental arithmetical test. The total possible mark was 10. The results were as follows. 4,8,5,2,3,8,7,6,6,4,2,5,8,7,6,6. Inter these results into a suitable frequency distribution table

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(b) In ΔABC M is on AB with $AM = 4cm$ and $MB = 2cm$ N is on AC and MN is parallel to BC



(i) Show that ΔAMN is similar to ΔABC

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(ii) If $BC = 4.5cm$ find MN

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9. (a) Musa walks $1km$ up the side of the hill. If he has climbed vertical distance of $600m$. Find the horizontal distance through which he has travelled.

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(b) There are 25 students in a class. 11 Play rugby 16 play football and 5 play neither. Find how many both play.

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10. (a) Find the vector of translation which take $(1,2)\ to\ (4,7)$

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(b) Solve the following quadratic equation by using the general formula $9x^2 + 9x - 54 = 0$

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(c) If $q * q = \frac{p-(p-q)}{p}$ find $2 * 4$

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